University of Calgary
Faculty of Science Final Exam
PHYSICS 365

December 21, 2022, 12:00 PM – 3:00 PM Time: 180 minutes

This is a closed-book exam worth a total of 35 points. Please answer all questions.
ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the multiple-choice exam questions and the formula sheet (last 3 pages). THIS BOOKLET WILL **NOT** BE HANDED IN. Make sure this booklet contains 13 pages (including formula sheet). If you are missing any pages, get a new booklet from the exam supervisor.

2. A BUBBLE SHEET used to answer multiple-choice questions. **IT WILL BE HANDED IN**.

IMPORTANT: START by entering your ID NUMBER, NAME and COURSE ID on the bubble sheet. Using a pen or a pencil, black out the corresponding numbers below your ID and name. DO THIS NOW. Then wait for the Exam Supervisor to signal when to start the test. All answers to the multiple-choice questions must be entered by blacking out the appropriate character (one of A, B, C, D) beside the question number on the bubble sheet. Make sure you darken the entire interior of the circle that contains the character.

3. A PEN OR PENCIL, used to black out your answers on the bubble sheet. If you are using a PEN, make sure it is BLACK or BLUE in order to ensure the scanner reads your answers properly. PENCILS can also be used, but make sure you press firmly when answering so the scanner reads your answer properly.

4. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

Please note that you can leave the room at any time after the first 30 minutes (after the signing-in process have been completed) Students remaining during the *last 15 minutes* must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

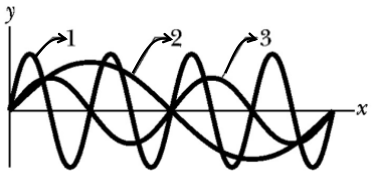
1. The following four waves are sent along strings with the same linear densities (x is in meters and t is in seconds). Rank the tension (τ) in the strings along which they travel, greatest first:

$$y_1 = (3\text{ mm})\sin(x - 3t)$$
$$y_2 = (6\text{ mm})\sin(2x - t)$$
$$y_3 = (1\text{ mm})\sin(4x - t)$$
$$y_4 = (2\text{ mm})\sin(x - 2t)$$

A) $\tau_1 > \tau_4 = \tau_2 > \tau_3$
B) $\tau_3 > \tau_2 = \tau_4 > \tau_1$
C) $\tau_3 > \tau_2 > \tau_4 > \tau_1$
D) $\tau_1 > \tau_4 > \tau_2 > \tau_3$

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2. The figure is a composite of three snapshots, each of a wave traveling along a particular string. The phases for the waves are given by **(a) 2x − 4t, (b) 4x − 8t**, and **(c) 8x − 16t.** Which phase corresponds to which wave in the figure?

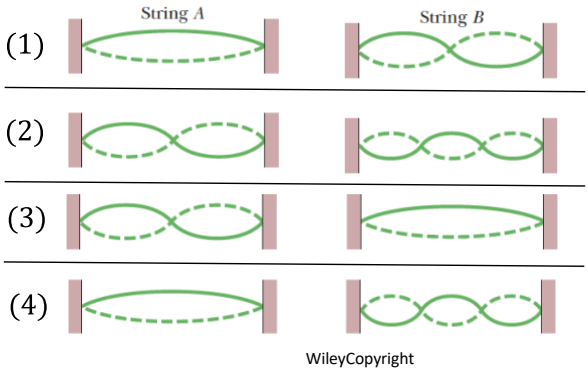


A) (a) to 1; (b) to 2; (c) to 3
B) (a) to 3; (b) to 2; (c) to 1
C) (a) to 2; (b) to 3; (c) to 1
D) (a) to 3; (b) to 1; (c) to 2

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3. Strings A and B have identical lengths and linear densities, but string B is under greater tension than string A. Figure shows four situations, (1) through (4), in which standing wave patterns exist on the two strings. In which situations is there the possibility that strings A and B are oscillating at the same resonant frequency?

A) (1)
B) (2)
C) (3)
D) (4)



4. Four waves are to be sent along the same string, in the same direction:

$y_1(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t)$
$y_2(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + 0.5\pi)$
$y_3(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + \pi)$
$y_4(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + 1.5\pi)$

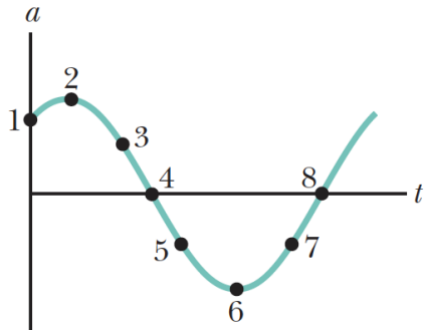
What is the wave function of the resultant wave of these waves?

A) $y'(x,t) = 0$
B) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t + 0.5\pi)$
C) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t)$
D) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t + \pi)$

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5. The acceleration $a(t)$ of a particle undergoing SHM with amplitude x_m is graphed in the figure. At point 5, particle is at:

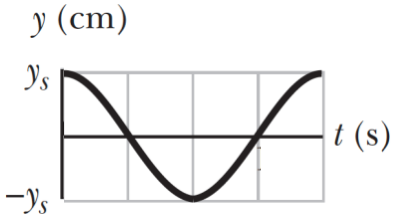
A) $+x_m$
B) $-x_m$
C) Between $-x_m$ and 0
D) Between 0 and $+x_m$



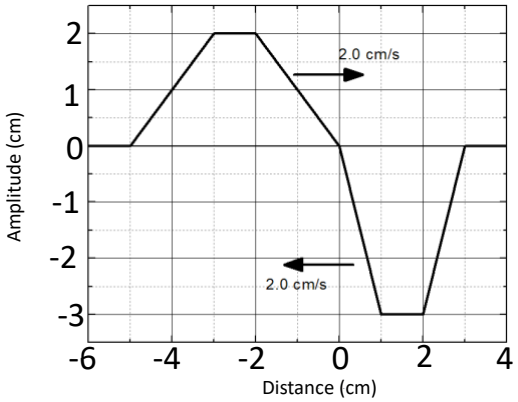
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6. A sinusoidal transverse wave travels along a string in the positive direction of the x axis. The wave equation form is $y(x,t) = y_s \sin(kx - \omega t + \varphi)$. The displacement y of the string particle at $x = 0$ is given in figure as a function of time t. The wave function is to be in the form of:

A) $y(x,t) = +y_s \sin(kx - \omega t)$
B) $y(x,t) = -y_s \sin(kx - \omega t)$
C) $y(x,t) = +y_s \cos(kx - \omega t)$
D) $y(x,t) = -y_s \cos(kx - \omega t)$



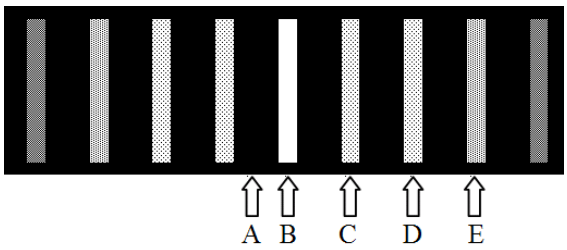
7. The graph shows two waves at time t = 0 s, one moving toward the right at 2.0 cm/s and the other moving toward the left at 2.0 cm/s. What will the amplitude be at x = 0 at time t = 1.0 s?



A) −3 cm
B) zero cm
C) −1 cm
D) −2 cm

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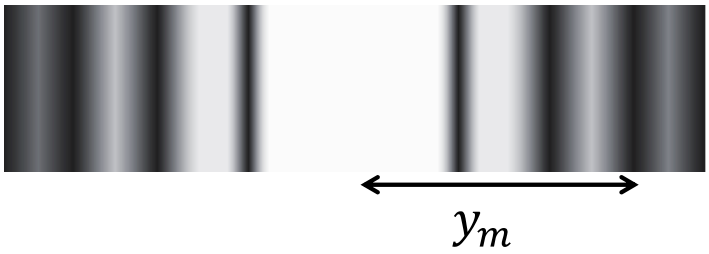
8. A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled "**B.**" Which one of these fringes is produced when the path length difference is 1980 nm?



A) A
B) C
C) D
D) E

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9. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2m from the slit. Determine the distance y_m.



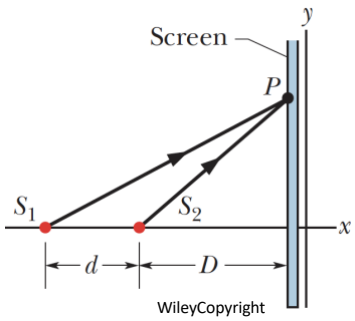
A) 0.003 m
B) 0.006 m.
C) 0.009 m
D) 0.012 m

Questions 10 and 11 are related to the following situation:

In the Figure, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 6.00\,\lambda$ on an x axis. A viewing screen is at distance $D = 20.0\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P.

10. What is the path length difference between two rays when $y_p = d$?

A) 3.50 λ
B) 20.0 λ
C) 5.80 λ
D) 26.7 λ



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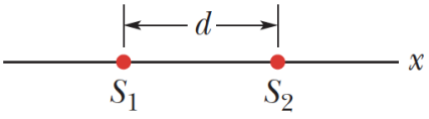
11. If the path length difference between two rays from S_1 and S_2 at P (above figure) equals to 7.4 λ, the resulting intensity at point P is

A) Minimum
B) Maximum
C) Intermediate but closer to minimum
D) Path length difference 7.4 λ for the above situation is impossible

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12. In the Figure, two radiofrequency point sources S_1 and S_2, separated by distance $d = 2.0$ m, are radiating in phase with $\lambda = 0.50$ m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

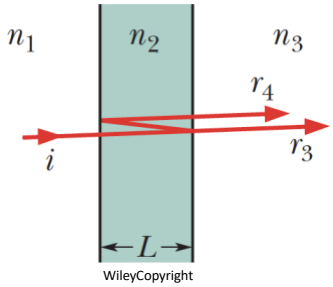
A) 9
B) 16
C) 18
D) 20



Questions 13 and 14 are related to the following situation:

In the Figure, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3(the light does not reflect inside material 2) and r_4(the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.

Question 13 find the wavelength that is in the visible range. (The visible wavelengths range from 0.4μm to 0.7μm)



Question	n_1	n_2	n_3	Type	L	λ
13.	1.32	1.75	1.39	min	335	

A) 498 nm
B) 480 nm
C) 469 nm
D) 455 nm

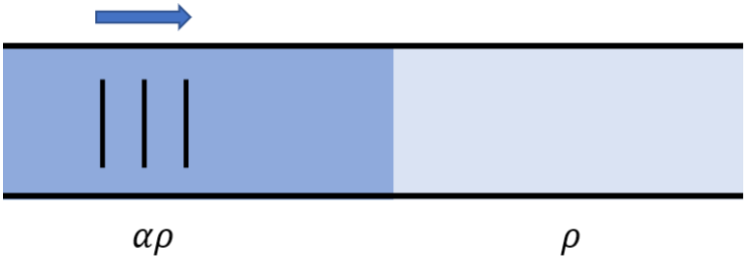
Question 14 find the third least thickness.

Question	n_1	n_2	n_3	Type	L	λ
14.	1.55	1.60	1.33	Min	3rd	512

A) 400 nm
B) 423 nm
C) 478 nm
D) 497 nm

15. A wave in a fluid is incident on a boundary with another fluid. The wave travels from the left, in a fluid with mass density $\alpha\rho$, to the right, into the fluid with mass density ρ (i.e., the mass density of the fluid on the left is α times that of the fluid on the right). The two fluids have the same bulk modulus. Which option gives *power* transmission coefficients (T) at this boundary?

A) $\frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}}$
B) $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$
C) $\frac{2}{1+\sqrt{\alpha}}$
D) $\frac{4\sqrt{\alpha}}{1+\sqrt{\alpha}}$



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16. A whistle of frequency 540 Hz moves in a circle of radius 60 cm at an angular speed of 15.0 rad/s. What is the *lowest* frequency heard by a listener a long distance away, at rest with respect to the center of the circle? (For a circular motion the linear velocity $v = r\omega$, whereas r is the radius of the circle and ω is the angular speed.)

A) 526 Hz
B) 534 Hz
C) 555 Hz
D) 565 Hz

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17. Which of the following statements is FALSE about the photoelectric effect?

A) The photoelectric effect is a consequence of the particle character of radiation.
B) There exists a minimum frequency of incident radiation below which photocurrent does not occur.
C) The work function defines the energy needed to detach a photoelectron from the surface of the target material.
D) There exists a lag time where electrons gain sufficient energy to leave the electrode surface.

18. What is a black body?

A) An object painted with black, non-reflective paint
B) An object that does not radiate any electromagnetic waves
C) A model of an object that emits electromagnetic waves with a characteristic frequency distribution
D) All of the above

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19. In the Compton effect, at what scattering angle, θ would the shift in the wavelength (i.e., Compton shift) become negligible.

A) $\theta = 90^o$
B) $\theta = 45^o$
C) $\theta = 0^o$
D) $\theta = 15^o$

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20. When the Compton shift was measured, what was demonstrated?

A) Photons have momentum.
B) Photons have energy.
C) Photons behave like waves.
D) Photons have no mass.

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21. If the frequency of vibration of an atom is f, the atom's energy is

A) Zero
B) 0.5 hf
C) 1.0 hf
D) 1.5 hf

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22. If the accuracy in measuring the position of a particle increases, the accuracy in measuring its momentum will:

A) increase.
B) decrease.
C) remain the same.
D) It is impossible to say since the two measurements are independent and do not affect each other

23. When the photoelectric effect experiments were performed, one effect was inconsistent with classical physics. What was it?

A) The fact that electrons could form a current within a vacuum.
B) The kinetic energy of the ejected electrons increased as the frequency of light increased.
C) The fact that light could free electrons from the surface of a metal.
D) The kinetic energy of the ejected electrons did not vary with light intensity.

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24. Photon A has twice the energy of photon B. Which statement is true?

A) the momentum of A is greater than that of B and the wavelength of A is less than that of B
B) the momentum of A is less than that of B and the wavelength of A is greater than that of B
C) the momentum of A is greater than that of B and the wavelength of A is greater than that of B
D) the momentum of A is greater than that of B and the wavelength of A equals that of B

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25. Using Stefan's law, determine the total power of blackbody radiation emitted across the entire spectrum of wavelengths for a blackbody with the surface area of 25 m^2 at 60 degrees Celsius.

A) 4.72 W
B) 17.43 kW
C) 18.37 W
D) 7.45 kW

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26. What is the magnitude of the momentum associated with a photon having an energy equal to the electron rest energy?

A) $4.73\times 10^{-12}\,m$
B) $2.43\times 10^{-12}\,m$
C) $3.23\times 10^{-12}\,m$
D) $5.12\times 10^{-12}\,m$

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27. The position along the *x* axis of an electron is known to be between – 0.31 nm and + 0.31 nm. How would the uncertainty in the momentum of the electron change if the electron were allowed to be between –0.62 nm and +0.62 nm?

A) The uncertainty in the momentum would be twice its previous value.
B) The uncertainty in the momentum would be half of its previous value.
C) The uncertainty in the momentum would be four times its previous value.
D) The uncertainty in the momentum would be one fourth its previous value.

28. What percentage increase in wavelength leads to a 75% loss of photon energy in a photon-free electron collision?
    A) 200 %
    B) 300 %
    C) 400 %
    D) 500 %

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29. If you double the kinetic energy of a nonrelativistic particle, how does its de Broglie wavelength change?

    A) $1/2$
    B) 2
    C) $\sqrt{2}$
    D) $1/\sqrt{2}$

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30. The wave functions for an electron in an infinite, one dimensional potential well with length L along an x axis is given by $\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi}{L}x\right)$. What is the probability of finding the electron between $x = (0.50)L$ and $x + dx = (0.51)L$?

    A) 0.005
    B) 0.014
    C) 0.020
    D) 0.050

PHYS 365 Formula Sheet

OSCILLATIONS			
$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{L}}$	
WAVES			
$y(x,t) = f(x \pm vt)$	$y(x,t) = A\sin(kx \pm \omega t + \phi_0)$		
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_{vacuum}}{n}$
$v = \sqrt{\frac{F_T}{\mu}}$	$v = \sqrt{\frac{B}{\rho}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{\gamma RT}{M}}$
$Z = \sqrt{\mu F_T} = \frac{F_T}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$p_{max} = BkA = Zv_{max}$	$I = \frac{p_{max}^2}{2\rho v}$
$P = \omega^2 A^2 Z\cos^2(\omega t \pm kx + \phi_0)$	$P_{av} = \frac{1}{2}\omega^2 A^2 Z$	$I_{avg} = \frac{1}{2}\omega^2 A^2 Z$	$I = \frac{P}{4\pi R^2}$
$\beta = (10dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\text{dB})\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$t = \frac{2Z_1}{Z_1 + Z_2}$	$r^2 + \frac{Z_2}{Z_1}t^2 = 1$	$Y = \frac{(\text{F/A})}{(\Delta L/L)}$
$\frac{\partial y}{\partial t} = \mp v\frac{\partial y}{\partial x}$	$\frac{\partial^2 y}{\partial^2 t} = v^2\frac{\partial^2 y}{\partial x^2}$	$\Delta p = -B\frac{\partial s}{\partial x}$	$B = -\frac{\Delta p}{(\Delta V/V)}$
$y(x,t) = 2A\sin(kx)\cos(\omega t)$		$f_n = \frac{v}{\lambda_n} = nf_1$	$f_L = \frac{v \pm v_L}{v \mp v_S} f_S$
$\Delta = \lvert r_2 - r_1\rvert = m\lambda$	$d\sin\theta = m\lambda$	$\Delta = a\sin\theta = m\lambda$	
MATHEMATICAL FORMULAE			
$\cos\alpha + \cos\beta = 2\cos\left[\frac{\alpha+\beta}{2}\right]\cos\left[\frac{\alpha-\beta}{2}\right]$		$\tan^{-1}(x) = \{\theta, \theta + \pi\} + 2\pi n$	
$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$		$\cos^{-1}(x) = \pm\theta + 2\pi n$	
		$\sin^{-1}(x) = \{\theta, \pi - \theta\} + 2\pi n$	
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$	$\tilde{A} = Ae^{j\theta} = A(\cos\theta + j\sin\theta)$	
CONSTANTS			
$c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$	$R = 8.314 \frac{\text{J}}{\text{K}\cdot\text{mol}}$	$M_{air} = 28.8 \frac{\text{g}}{\text{mol}}$	$g = 9.81 \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20°\text{C}} = 343 \frac{\text{m}}{\text{s}}$	$T_K = T_{°\text{C}} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$	$1\text{atm} = 1.01 \times 10^5\text{Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	

$$R = r^2, \quad T = \frac{Z_2}{Z_1}t^2$$

MODERN PHYSICS			
$F_e = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$	$U = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$	$U = qV$	$K = \frac{1}{2}mv^2$
$E = hf$	$p = \frac{h}{\lambda}$	$hf = K_{max} + \Phi$	$a_c = \frac{v^2}{r}$
$\Delta\lambda = \frac{h}{mc}(1 - \cos\varphi)$	$\frac{1}{\lambda} = R\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$	$R = \frac{P}{hf}$	
$\Delta x \Delta p_x \geq \frac{h}{2\pi}$			
$a = \varepsilon_0\frac{n^2h^2}{\pi m_e e^2}$	$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2}$	$E_n = -13.6\frac{\text{eV}}{n^2}$	
$E_n = \frac{h^2}{8m_e L^2}n^2$	$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$	$T \approx e^{-2bL}$, $b = \sqrt{8\pi^2 m(U_b - E)/h^2}$	
$P(r) = \frac{4}{a_0^3}r^2 e^{-\frac{2r}{a_0}}$	$\psi(r) = \frac{1}{\sqrt{\pi}a^{\frac{3}{2}}}e^{-\frac{r}{a}}$	$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E - U(x))\psi = 0$	
$\frac{d^2\psi}{dx^2} + k^2\psi = 0$, with $k = \frac{2\pi\sqrt{2m(E-U)}}{h}$			

MATHEMATICAL FORMULAE	
$\int \cos(ax) = \frac{\sin(ax)}{a} + c$	$\int \sin(ax) = -\frac{\cos(ax)}{a} + c$
$\int \sin^2(ax)dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + c$	$\int \cos^2(ax)dx = \frac{x}{2} + \frac{\sin(2ax)}{4a} + c$
$\sin^2\theta = \frac{1}{2}(1 - \cos(2\theta))$	$\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$

CONSTANTS	
$m_e = 9.109 \cdot 10^{-31}$ kg	$m_p = 1.67 \cdot 10^{-27}$ kg
$a_0 = 0.5292 \cdot 10^{-10}$ m	$1\,u = 1.66054 \cdot 10^{-27}$ kg
$\varepsilon_0 = 8.854 \cdot 10^{-12}\frac{\text{C}^2}{\text{Nm}^2}$	$h = 6.63 \cdot 10^{-34}\,\text{J}\cdot\text{s} = 4.136 \cdot 10^{-15}\text{eV}\cdot\text{s}$
$\sigma = 5.67 \cdot 10^{-8}\frac{\text{W}}{\text{m}^2\text{K}^2}$	$e = 1.602 \cdot 10^{-19}$ C
$k_B = 1.381 \cdot 10^{-23}\frac{\text{J}}{\text{K}} = 8.617 \cdot 10^{-5}\frac{\text{eV}}{\text{K}}$	$R = 1.09737 \cdot 10^{7}\,\text{m}^{-1}$

$$P = IA = \sigma T^4 A$$

$$p(x) = \psi_n^2(x)\,dx$$

$$c = 3.0 \times 10^8 m/s$$

$$E = pc \ \ (photon)$$

$$x = x_m \cos(\omega t + \phi) \quad \text{(displacement)}$$
$$v = -\omega x_m \sin(\omega t + \phi) \quad \text{(velocity)}$$
$$a = -\omega^2 x_m \cos(\omega t + \phi) \quad \text{(acceleration)}$$

$$\omega = \sqrt{\frac{k}{m}} \quad \text{(angular frequency)}$$

$$T = 2\pi\sqrt{\frac{m}{k}} \quad \text{(period).}$$

$$y(x, t) = y_m \sin(kx \pm \omega t + \phi)$$

$$k = \frac{2\pi}{\lambda} \qquad \frac{\omega}{2\pi} = f = \frac{1}{T} \qquad v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f.$$

$$P_{\text{avg}} = \tfrac{1}{2}\mu v \omega^2 y_m^2 \qquad v = \sqrt{\frac{\tau}{\mu}}.$$

$$y'(x, t) = [2y_m \cos \tfrac{1}{2}\phi] \sin(kx - \omega t + \tfrac{1}{2}\phi)$$

$$y'(x, t) = [2y_m \sin kx] \cos \omega t.$$

$$f_{\text{beat}} = f_1 - f_2$$

$$f' = f\frac{v \pm v_D}{v \pm v_S} \quad \text{(general Doppler effect)}$$

$$d \sin\theta = m\lambda, \qquad \text{for } m = 0, 1, 2, \ldots$$
(maxima — bright fringes),

$$d \sin\theta = (m + \tfrac{1}{2})\lambda, \qquad \text{for } m = 0, 1, 2, \ldots$$
(minima — dark fringes),

Single-Slit Diffraction

$$a \sin\theta = m\lambda, \qquad \text{for } m = 1, 2, 3, \ldots \quad \text{(minima).}$$

$$s = s_m \cos(kx - \omega t),$$

$$\Delta p = \Delta p_m \sin(kx - \omega t)$$

$$\Delta p_m = (v\rho\omega)s_m$$

$$\phi = \frac{\Delta L}{\lambda} 2\pi$$

constructive interference

$$\frac{\Delta L}{\lambda} = 0, 1, 2, \ldots.$$

destructive interference

$$\frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \ldots.$$

$$I = \frac{P}{A}$$

$$I = \tfrac{1}{2}\rho v \omega^2 s_m^2 \qquad I = \frac{P_s}{4\pi r^2}$$

$$\beta = (10 \text{ dB}) \log \frac{I}{I_0} \quad I_0 \ (= 10^{-12} \text{ W/m}^2)$$

pipe open at both ends

$$f = \frac{v}{\lambda} = \frac{nv}{2L}, \quad n = 1, 2, 3, \ldots,$$

pipe closed at one end and open at the other

$$f = \frac{v}{\lambda} = \frac{nv}{4L}, \quad n = 1, 3, 5, \ldots.$$

Intensity in Two-Slit Interference

$$I = 4I_0 \cos^2 \tfrac{1}{2}\phi, \qquad \text{where } \phi = \frac{2\pi d}{\lambda} \sin\theta$$

Reflection	Reflection phase shift
Off lower index	0
Off higher index	0.5 wavelength

$$\sin(-\alpha) = -\sin\alpha$$
$$\cos(-\alpha) = \cos\alpha$$
$$\sin(\alpha \pm \pi/2) = \pm\cos\alpha$$
$$\cos(\alpha \pm \pi/2) = \mp\sin\alpha$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \pm \sin\alpha\sin\beta$$
$$\sin\alpha + \sin\beta = 2\sin\tfrac{1}{2}(\alpha + \beta)\cos\tfrac{1}{2}(\alpha - \beta)$$
$$\cos\alpha + \cos\beta = 2\cos\tfrac{1}{2}(\alpha + \beta)\cos\tfrac{1}{2}(\alpha - \beta)$$